

# Assignment 03 — Solutions

CSE 350 — Spring 2026

RTL &amp; DTL gate analysis — power, fanout, noise margin (Total: 30 marks)

## Question 1 — RTL Inverter with 48 loads

### Given

$V_{OH} = 4.5 \text{ V}$ ,  $V_{OL} = 0.2 \text{ V}$ ,  $V_{CC} = 5 \text{ V}$ ,  $V_R = -5 \text{ V}$   
 $R_{in} = 5 \text{ k}\Omega$ ,  $R_B = 100 \text{ k}\Omega$ ,  $R_C = 2 \text{ k}\Omega$   
 $\beta_F = 30$ ,  $V_{BE(sat)} = 0.8 \text{ V}$ ,  $V_{CE(sat)} = 0.2 \text{ V}$ ,  $V_Y = 0.5 \text{ V}$   
 Driver and loads are identical;  $N = 48$  loads connected.

### (a) Power dissipation ( $V_{in} = \text{HIGH}$ , 48 loads)

Driver base currents when  $V_{in} = V_{OH} = 4.5 \text{ V}$ :

$$I_{I1} = (V_{OH} - V_{BE}) / R_{in} = (4.5 - 0.8) / 5\text{k} = 0.74 \text{ mA}$$

$$I_{I2} = (V_{BE} - V_R) / R_B = (0.8 - (-5)) / 100\text{k} = 0.058 \text{ mA}$$

$$I_B = I_{I1} - I_{I2} = 0.682 \text{ mA}$$

Each load is OFF (its input sees  $V_{OL} = 0.2 \text{ V}$  from the driver). The load base voltage is set by the divider between  $V_{OL}$  and  $V_R$ :

$$\begin{aligned}
 V_{B,load} &= (V_{OL} \cdot R_B + V_R \cdot R_{in}) / (R_{in} + R_B) \\
 &= (0.2 \cdot 100 + (-5) \cdot 5) / 105 = -0.0476 \text{ V}
 \end{aligned}$$

$V_{B,load} \ll V_Y = 0.5 \text{ V} \rightarrow$  loads are OFF ✓. Current drawn into the driver output from each load:

$$I_{IL} = (V_{OL} - V_{B,load}) / R_{in} = (0.2 + 0.0476) / 5\text{k} = 0.0495 \text{ mA}$$

Driver's own  $R_C$  contribution and total collector current:

$$I_{RC} = (V_{CC} - V_{CE}) / R_C = (5 - 0.2) / 2\text{k} = 2.4 \text{ mA}$$

$$I_C = I_{RC} + 48 \cdot I_{IL} = 2.4 + 2.376 = 4.776 \text{ mA}$$

Saturation check:  $\beta \cdot I_B = 30 \times 0.682 = 20.46 \text{ mA} \gg 4.776 \text{ mA}$  ✓

Power supplied by the driver's own sources ( $V_{CC}$ ,  $V_{in}$ ,  $|V_R|$ ):

$$P_{VCC} = 5 \times 2.4 = 12.00 \text{ mW} \quad (R_C \text{ path})$$

$$P_{Vin} = 4.5 \times 0.74 = 3.33 \text{ mW} \quad (R_{in} \text{ path})$$

$$P_{VR} = 5 \times 0.058 = 0.29 \text{ mW} \quad (R_B \text{ path to } V_R)$$

$$P_D = 15.62 \text{ mW}$$

(The 2.376 mA flowing into the driver's collector from the 48 loads is supplied by the load gates'  $V_{CC}$  rails, so it does not count toward the driver's own dissipation.)

**Answer:**  $P_D \approx 15.62 \text{ mW}$

### (b) Noise margin

$V_{IL}$  — maximum input voltage that keeps the driver OFF ( $V_{B,driver} = V_Y$  at the boundary):

$$V_Y = (V_{IL} \cdot R_B + V_R \cdot R_{in}) / (R_{in} + R_B)$$

$$0.5 = (100 \cdot V_{IL} - 25) / 105$$

$$100 \cdot V_{IL} = 52.5 + 25 = 77.5$$

$$V_{IL} = 0.775 \text{ V}$$

$V_{IH}$  — minimum input that keeps driver at the edge of saturation (own  $R_C$  only). Required base current:

$$I_{B,min} = I_{R_C} / \beta_F = 2.4 / 30 = 0.08 \text{ mA}$$

$$I_{B,min} = (V_{IH} - V_{BE})/R_{in} - (V_{BE} - V_R)/R_B$$

$$0.08 = (V_{IH} - 0.8)/5 - 0.058$$

$$(V_{IH} - 0.8)/5 = 0.138$$

$$V_{IH} = 1.49 \text{ V}$$

Noise margins:

$$NM_L = V_{IL} - V_{OL} = 0.775 - 0.2 = 0.575 \text{ V}$$

$$NM_H = V_{OH} - V_{IH} = 4.5 - 1.49 = 3.01 \text{ V}$$

**Answer:**  $NM_L = 0.575 \text{ V}$ ,  $NM_H = 3.01 \text{ V}$

## Question 2 — DTL NAND

### Given

$$V_{CC} = 10 \text{ V}, \quad V_{OH} = 3.5 \text{ V}, \quad V_{OL} = 0.1 \text{ V}, \quad \beta_F = 30$$

Driver:  $R_1 = 2 \text{ k}\Omega$  (input pull-up),  $R_2 = 5 \text{ k}\Omega$  (Q1 base→GND),  $R_3 = 6 \text{ k}\Omega$  (Q1 collector pull-up)

Loads:  $R_4 = 2 \text{ k}\Omega$  (load input pull-up),  $R_5 = 5 \text{ k}\Omega$ ,  $R_6 = 6 \text{ k}\Omega$  — identical to driver

$$\text{Assumed: } V_{BE}(\text{sat}) = 0.8 \text{ V}, \quad V_{CE}(\text{sat}) = V_{OL} = 0.1 \text{ V}, \quad V_D = 0.7 \text{ V}$$

### (a) Maximum fanout

DTL NAND output is LOW when all inputs are HIGH ( $V_{OH} = 3.5 \text{ V}$ ). With all inputs HIGH, the input diodes are reverse-biased ( $V_{\text{input}} > V_P$ ), so current flows  $V_{CC} \rightarrow R_1 \rightarrow P \rightarrow D_4 \rightarrow D_5 \rightarrow \text{base of T1} \rightarrow \text{ground}$ , turning T1 saturated.

$$V_P = V_{BE}(\text{sat}) + 2 \cdot V_D = 0.8 + 1.4 = 2.2 \text{ V}$$

Diode reverse-bias check:  $V_{\text{input}} - V_P = 3.5 - 2.2 = 1.3 \text{ V}$  (cathode positive)  $\rightarrow$  reverse-biased ✓

$$I_1 = (V_{CC} - V_P) / R_1 = (10 - 2.2) / 2\text{k} = 3.9 \text{ mA}$$

$$I_2 = V_{BE}(\text{sat}) / R_2 = 0.8 / 5\text{k} = 0.16 \text{ mA}$$

$$I_B = I_1 - I_2 = 3.74 \text{ mA}$$

Driver's own  $R_3$  (collector pull-up):

$$I_{R_3} = (V_{CC} - V_{CE}) / R_3 = (10 - 0.1) / 6\text{k} = 1.65 \text{ mA}$$

Each load's input diode conducts when driver  $Y = 0.1 \text{ V}$  (load  $V_{P,\text{load}} = 0.1 + 0.7 = 0.8 \text{ V}$ ). Current from the load's  $V_{CC}$  sourced through its  $R_4$  into the driver output:

$$I_L = (V_{CC} - V_{P,\text{load}}) / R_4 = (10 - 0.8) / 2\text{k} = 4.6 \text{ mA per load}$$

Saturation condition  $I_{R_3} + N \cdot I_L \leq \beta_F \cdot I_B$ :

$$1.65 + 4.6 \cdot N \leq 30 \times 3.74 = 112.2$$

$$N \leq 110.55 / 4.6 = 24.03$$

**Answer:  $N_{\text{max}} = 24$**

### (b) Maximum power dissipation, $N = 24$ loads

With  $N = 24$  loads, the driver's transistor is sinking  $I_{R_3} + 24 \cdot I_L = 112.05 \text{ mA}$  in total. However, the  $24 \cdot I_L = 110.4 \text{ mA}$  is sourced by the 24 load gates'  $V_{CC}$  rails — not by the driver's own  $V_{CC}$ .

The driver's own  $V_{CC}$  supplies two paths:  $I_1$  (through  $R_1$ ) and  $I_{R_3}$  (through  $R_3$ ).

$$\text{Total current from driver } V_{CC} = I_1 + I_{R_3} = 3.9 + 1.65 = 5.55 \text{ mA}$$

$$P_D = V_{CC} \times 5.55 = 10 \times 5.55 = 55.5 \text{ mW}$$

**Answer:  $P_D = 55.5 \text{ mW}$**

## Question 3 — RTL with mixed driver/load topology

### Given

$V_{CC} = 12\text{ V}$ ,  $V_{OH} = 10\text{ V}$ ,  $V_{OL} = 0.2\text{ V}$ , high input = 12 V, low input = 0.2 V  
 Driver Q<sub>D</sub>:  $R_{in} = 15\text{ k}\Omega$ ,  $R_C = 2.2\text{ k}\Omega$ ,  $R_B = 100\text{ k}\Omega \rightarrow V_R = -12\text{ V}$   
 Load Q<sub>L</sub>:  $R_{in} = 20\text{ k}\Omega$ ,  $R_C = 2\text{ k}\Omega$ ,  $R_B = 90\text{ k}\Omega \rightarrow \text{GND}$   
 $\beta_F = 25$ ,  $V_{BE}(\text{sat}) = 0.8\text{ V}$ ,  $V_{CE}(\text{sat}) = 0.2\text{ V}$ ,  $V_Y = 0.5\text{ V}$  (assumed)

### (a) Noise margin

$V_{IL}$  — max input keeping the driver OFF ( $V_{B,Q_D} = V_Y$  at boundary):

$$\begin{aligned}
 V_Y &= (V_{IL} \cdot R_B + V_R \cdot R_{in}) / (R_{in} + R_B) \\
 0.5 &= (100 \cdot V_{IL} + (-12)(15)) / 115 = (100 \cdot V_{IL} - 180) / 115 \\
 57.5 + 180 &= 100 \cdot V_{IL} \rightarrow V_{IL} = 2.375\text{ V}
 \end{aligned}$$

$V_{IH}$  — min input keeping the driver at the edge of saturation. Required base current with no loads counted:

$$\begin{aligned}
 I_{R_C} &= (12 - 0.2) / 2.2\text{ k} = 5.364\text{ mA} \\
 I_{B,\min} &= I_{R_C} / \beta_F = 5.364 / 25 = 0.2146\text{ mA} \\
 I_2 &= (V_{BE} - V_R) / R_B = (0.8 + 12) / 100\text{ k} = 0.128\text{ mA} \\
 I_{B,\min} &= (V_{IH} - V_{BE}) / R_{in} - I_2 \\
 0.2146 &= (V_{IH} - 0.8) / 15 - 0.128 \\
 (V_{IH} - 0.8) / 15 &= 0.3426 \rightarrow V_{IH} = 5.94\text{ V}
 \end{aligned}$$

Noise margins (using the given input swing 0.2 V  $\rightarrow$  12 V):

$$\begin{aligned}
 NM_L &= V_{IL} - V_{in,\text{low}} = 2.375 - 0.2 = 2.175\text{ V} \\
 NM_H &= V_{in,\text{high}} - V_{IH} = 12 - 5.94 = 6.06\text{ V}
 \end{aligned}$$

**Answer:**  $NM_L = 2.175\text{ V}$ ,  $NM_H = 6.06\text{ V}$

**Note:** If the convention is to use  $V_{OH}$  (= 10 V) as the high input level rather than the stated 12 V,  $NM_H = 10 - 5.94 = 4.06\text{ V}$ . The problem statement gives 'high input = 12 V' explicitly, so 6.06 V is the literal answer; verify which convention your instructor expects.

### (b) Maximum fanout

Driver ON,  $V_A = 12\text{ V}$  (given high input). Driver base currents:

$$\begin{aligned}
 I_1 &= (12 - 0.8) / 15\text{ k} = 0.7467\text{ mA} \\
 I_2 &= (0.8 + 12) / 100\text{ k} = 0.1280\text{ mA} \\
 I_B &= I_1 - I_2 = 0.6187\text{ mA} \\
 I_{C,\max} &= \beta_F \cdot I_B = 25 \times 0.6187 = 15.47\text{ mA}
 \end{aligned}$$

Driver's own  $R_C$ :  $I_{R_C} = 5.364\text{ mA}$  (as above).

Each load has a different topology ( $R_B = 90\text{ k}\Omega$  to GND, not to  $V_R$ ). When driver  $Y = V_{OL} = 0.2\text{ V}$ , the load base voltage is just the divider:

$$V_{B,\text{load}} = V_{OL} \cdot 90 / (20 + 90) = 0.2 \times 90 / 110 = 0.1636\text{ V}$$

$V_{B,\text{load}} = 0.164\text{ V} \ll V_Y = 0.5\text{ V} \rightarrow$  load OFF ✓. Current drawn from the driver output by one load (just the series divider current):

$$I_L = V_{OL} / (R_{in,load} + R_{B,load}) = 0.2 / 110k = 1.818 \mu A$$

Fanout condition  $I_{R_C} + N \cdot I_L \leq I_{C,max}$ :

$$N \leq (15.47 - 5.364) \text{ mA} / 1.818 \mu A \approx 5560$$

**Answer:**  $N_{max} \approx 5560$  (DC current limit; effectively unlimited at DC)

**Note:** This number is unrealistically large because the load's input impedance (110 k $\Omega$ ) is far higher than the driver's  $R_C$  (2.2 k $\Omega$ ), so the driver can DC-sink the tiny load current trivially. In practice the fanout would be limited by signal integrity and propagation delay, not DC current. If your instructor expected a more standard answer, they may have intended a different load topology (e.g.,  $R_{B,load}$  also tied to  $V_R$ ) or different given values; the calculation above strictly follows the figure.

## Summary of Answers

| Q | Part | Answer                                                    |
|---|------|-----------------------------------------------------------|
| 1 | (a)  | $P_D \approx 15.62 \text{ mW}$                            |
| 1 | (b)  | $NM_L = 0.575 \text{ V}$ , $NM_H = 3.01 \text{ V}$        |
| 2 | (a)  | $N_{\max} = 24$                                           |
| 2 | (b)  | $P_D = 55.5 \text{ mW}$                                   |
| 3 | (a)  | $NM_L = 2.175 \text{ V}$ , $NM_H = 6.06 \text{ V}$        |
| 3 | (b)  | $N_{\max} \approx 5560$ (DC limit — see note in solution) |

### Caveats to verify with instructor

- Q1(a): Power dissipation is computed for  $V_{in} = \text{HIGH}$  (worst case — driver saturated). Counted are only the driver's own supplies ( $V_{CC}$ ,  $V_{in}$ ,  $|V_R|$ ). The 2.4 mA flowing into the driver's collector from the 48 loads is supplied by the load gates'  $V_{CC}$  rails.
- Q3(a): The high input is given explicitly as 12 V (not  $V_{OH} = 10 \text{ V}$ ).  $NM_H = 6.06 \text{ V}$  uses 12 V as written; using 10 V instead gives  $NM_H = 4.06 \text{ V}$ .
- Q3(b): The fanout result ( $\sim 5560$ ) is the DC current limit and is unrealistically large because the load's RTL input impedance ( $110 \text{ k}\Omega$ ) is far higher than the driver's  $R_C$  ( $2.2 \text{ k}\Omega$ ). If your instructor expected a smaller, more standard answer, the load topology in the figure may need re-examination.
- $V_y = 0.5 \text{ V}$  was assumed for Q3 since it was not stated; standard value used throughout.
- Diode forward drop  $V_D = 0.7 \text{ V}$  was assumed for Q2.